MILITARY SPECIFICATION

AIR CONDITIONER A/E32C-39

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for a 60/50-Hz, electric-motor-driven military air conditioner, designated A/E32C-39, having a nominal cooling capacity of 54,000 Btu/hr.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

F-F-300	Filter, Air Conditioning: Viscous-impingement and Dry Types, Cleanable
BB-F-1421	Fluorocarbon Refrigerants
QQ-B-654	Brazing Alloys, Silver
RR-C-271	Chains and Attachments, Welded and Weldless
TT-C-490	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings
WW-T-799	Tube, Copper, Seamless, Water and Refrigeration (For Use with Solder-Flared or Compression Type Fittings)
PPP-B-601	Box, Wood, Cleated-Plywood

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-V-173	Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment)

FSC 4120

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MIL-S-3950	Switch, Toggle, General Specification for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-P-23377	Primer Coating; Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-T-27260	Tie Down, Cargo Aircraft, CGU-1/B
MIL-D-38386	Duct Assemblies, Ground Conditioned Air, Insulated, Flexible
MIL-C-83286	Coating, Urethane, Aliphatic Isocyanate, for Aerospace Applications

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-471	Maintainability/Verification/Demonstration/Evaluation
MIL-STD-808	Finishes, Protective, and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods
MS90558	Connector, Receptacle, Electrical, Wall Mounting (with Coupling Ring) Class L (Equipment Receptacle)

DRAWING

AIR FORCE

72289-25	Cable Assembly - 120/208V-3 Phase, 4 Wire Grounding 60 Hz - 60 amps

U.S. Army Mobility Equipment Research and Development Center

13216E6168	Valve, Charging, with Cap

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society of Heating, Refrigerating, and Air Conditioning Engineers, Incorporated

ASHRAE Guide and Data Book

Standard 37-60 Methods of Testing for Rating Unitary Air Conditioning Equipment

(Application for copies should be addressed to the American Society of Heating, Refrigerating, and Air Conditioning Engineers, Incorporated; United Engineering Center; 345 East 47th Street; New York, NY 10017.)

Air Moving and Conditioning Association, Inc.

Bulletin 210 AMCA Standard Test Code for Air Moving Devices

(Application for copies should be addressed to the Air Moving and Conditioning Association, 205 West Touhy Avenue, Park Ridge, IL 60068.)

American National Standards Institute, Inc.

S1.4 - 1961 General Purpose Sound Level Meters

S1.11 - 1966 Octave, Half Octave, and Third Octave Base Filters Sets

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

Underwriters' Laboratories

UL 984 Standard for Safety Sealed (Hermetic Type) Motor Compressors

(Application for copies should be addressed to the Underwriters' Laboratories Inc., 207 E. Ohio St., Chicago, IL 60611.)

DRAWING

American Air Filter

DL 12090 Air Conditioner, A/E32C-39 4-1/2 Ton Nominal Capacity, 208 V/50/60 Hz/3 Phase.

(Application for copies should be addressed to American Air Filter, 1270 North Pine Road, St. Louis, MO 63132.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

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3.1.1 Drawings. The requirements of American Air Filter Drawing List DL 12090 apply as requirements of this specification with the exceptions and additions specified herein (see 6.2.b).

3.2 Components. The air conditioner shall consist of the following major components:

Description	See Requirement
Air system	3.7.1
Refrigeration system	3.7.2
Control system	3.7.3
Electrical system	3.7.4
Housing	3.7.5.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Fungusproof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.4.2.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.3 Gasketing and insulating materials. Where rubber or similar gasketing and insulating materials are used, they shall be compounded to insure their oil resistance and, where applicable, their ability to withstand low temperatures.

3.5 Design and construction

3.5.1 Reliability. The air conditioner shall have a minimum mean-time-between-failure (MTBF) of 434 hours at a confidence factor of 0.90 (see 6.3.1, 6.3.2, and 6.4).

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3.5.2 Configuration. The air conditioner shall be of the horizontal configuration. The evaporator inlet and outlet shall be provided with flanges to provide for attachment of free discharge grilles, or flexible duct adapters. Openings in the sides of the unit for condenser air circulation will be permissible providing such openings are confined to the rear half of the unit. Louvers or other protection shall be provided over all condenser air openings so that a 5/8-inch-diameter rod will not pass through. Two 4-inch by 12-inch nominal fork lift channels shall be provided in the side of the unit, on centers of not more than 48 inches. The unit shall be configured such that the supply and return ducts, the power cable and the drain hoses and fittings can be placed internally for storage and shipping without damage to them or components of the air conditioner. The air conditioner shall be designed and configured such that 9.5 KW, two-step electric strip heaters will be provided.

3.5.3 Electric power source. The air conditioner shall be designed to operate from a 120/208 volt, 50/60 Hz, 3-phase, 4-wire, grounding system delivering from 110/191 to 126/218 volts.

3.5.4 Acceleration loadings. The air conditioner shall be designed and constructed to withstand the acceleration forces specified in MIL-A-8421 for flight and taxiing and emergency landing loads when stacked three units high.

3.5.5 Maintainability. The air conditioner design and construction shall minimize maintenance time and the possibility of human failures. The following shall apply:

3.5.5.1 Accessibility. The air conditioner shall be so designed and constructed that the following actions may be accomplished without removing additional components or their subassemblies (other than panels) in not more than the time indicated: Maximum time to change out or repair a major component - 1 hour (exclusive of evacuation and recharging).

3.5.5.2 Clearances. To the maximum practicable extent, maintainability provisions shall incorporate features insuring operating clearances for facilitating maintenance and servicing.

3.5.5.3 Over-torquing. Intricate locking devices, controls, and threaded fastenings that can be easily over-torqued by personnel shall be avoided where possible.

3.5.5.4 Access openings. Covers or plates that must be removed for component adjustments or for component or part removal shall be equipped with quick-disconnect fastenings.

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3.5.5.5 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be mounted so that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.5.6 Field storage and transport. The air conditioner shall be designed and constructed to withstand damage from stacking the units four high for storage without shipping containers and three high for transportation without shipping containers, but with four cargo tiedowns conforming to MIL-T-27260.

3.6 Performance

3.6.1 Ambient operating range. The air conditioner shall be capable of starting and supplying refrigerated, recirculated air within the range of ambient conditions shown on figure 1. The capacity of the air conditioner shall be in accordance with 3.6.2.

3.6.1.1 Extreme operating conditions. The air conditioner shall have the capability to start and operate continuously in the cooling mode with air entering and being maintained to both the evaporator and condenser at 51.7°C (125°F). In addition, the air conditioner shall have the capability to start and operate continuously in the cooling mode under the conditions specified in 3.6.2.1.

3.6.2 Cooling capacity. The minimum cooling capacities at sea level of the preproduction and production air conditioners shall be as indicated in table I. The air conditioner shall have both duct adapters attached and shall be operating against 0-inch water gage static pressure during the capacity testing. The air conditioner power draw shall not exceed 10.5 KW while operating under condition 1 of table I.

3.6.2.1 Cooling capacity at 10°C (50°F) ambient. The cooling capacity of the air conditioner shall be demonstrated at condition 4 of table I. In addition, the minimum ambient temperature at which the air conditioner will operate in the cooling mode shall be established.

3.6.3 Evaporator airflow. When the air conditioner is operating under condition 1 of table I, the evaporator fan shall provide a minimum of 2200 cfm selectively adjusted ±20 percent at 0-inch water gage and a minimum of 2,000 cfm selectively adjusted ±20 percent against a static pressure of 1.0-inch water gage.

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TABLE I. Capacity rating conditions.

	Ambient Temp °C (°F)	Evaporator Inlet Temp °C (°F)		Production Minimum
Condition	Dry Bulb	Dry Bulb	Wet Bulb	(Btu/hr)
1	46.1 (115)	29.4 (85)	21.1 (70)	54,000
2	51.7 (125)	32.2 (90)	18.3 (65)	50,000
3	35.0 (95)	26.7 (80)	19.4 (67)	60,000
4	10.0 (50)	26.7 (80)	14.4 (58)	70,000

3.6.3.1 Fresh air. The air conditioner shall be capable of selectively supplying at least 200 cfm of filtered, cooled, fresh (outside) air when operating against an external static pressure of 1.0-inch water gage. The 200 cfm may be part of the airflow requirements of 3.6.3 when operating in the fresh air mode. The amount of fresh air shall be controlled by a manual damper. The damper actuator need not be located on the control panel.

3.6.4 Heating capacity. The strip heaters shall have a minimum heating capacity of 16,800 Btu/hr in the low heat mode and 31,000 Btu/hr in the high heat mode.

3.6.5 Dehumidification. The air conditioner shall demonstrate the capability of removing moisture from the conditioned air when tested in accordance with 4.6.3.1.3.

3.6.6 Environmental exposure. The air conditioner shall conform to all requirements specified herein after being exposed to the following environmental conditions:

3.6.6.1 Non-operating

a. Storage temperatures ranging from -62.2°C to 71.1°C (-80° to 160°F)

b. Relative humidity up to 100 percent, including conditions wherein condensation takes place in the form of water and frost

c. Altitudes corresponding to pressures ranging from 30 to 3.4 inches Hg (approximately an altitude of 50,000 feet)

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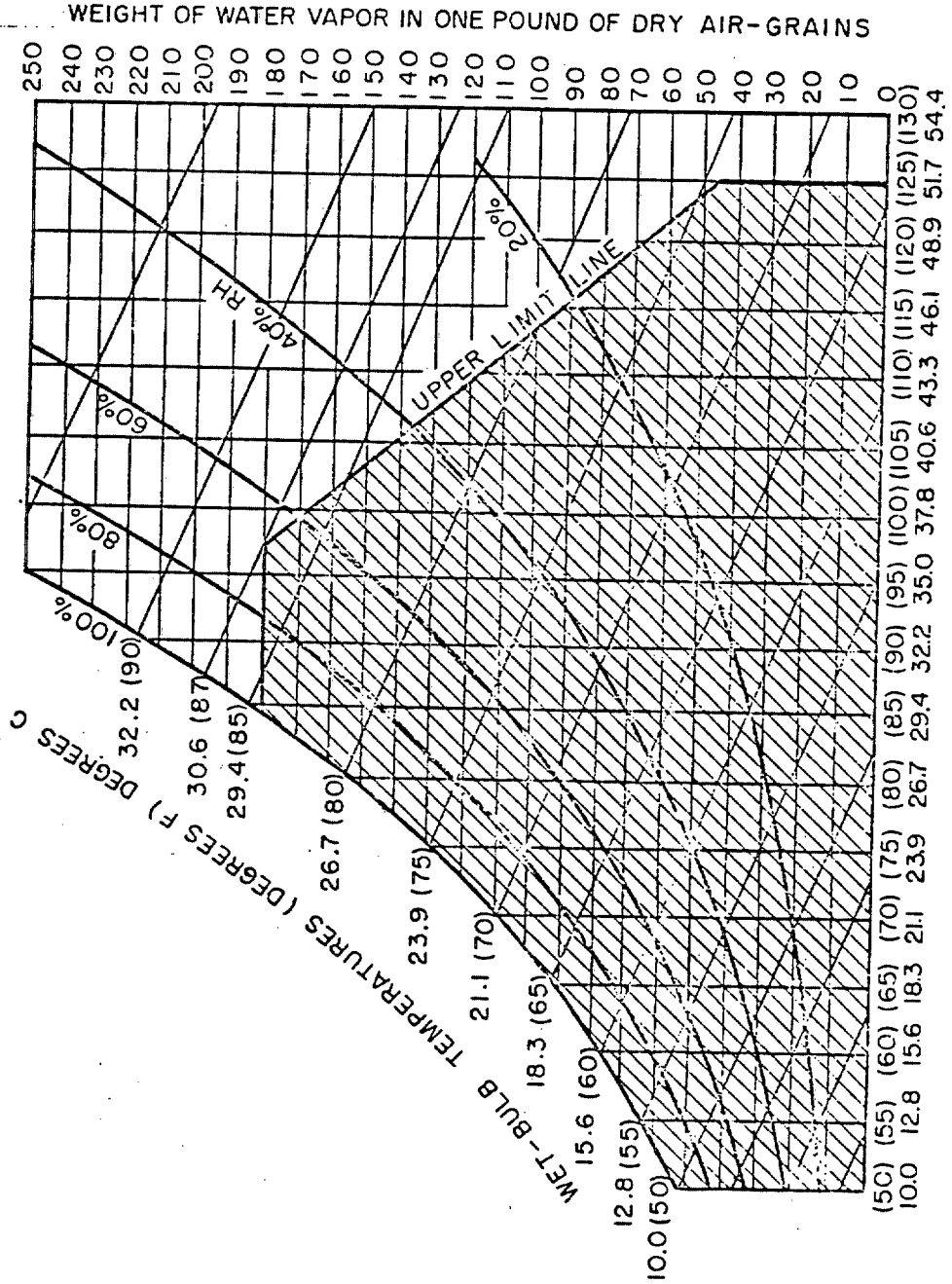


FIGURE 1. Ambient range for cooling operation.

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d. Exposure to salt-water sea atmosphere

e. Shock forces as anticipated in transport

f. Fungus growth as encountered in tropical climates

g. Radiant energy as found under natural conditions

h. Rainfall as encountered in any locale

i. Sand and dust particles as encountered in desert areas.

3.6.6.2 Operating

a. Shall operate satisfactorily at a 6,000 foot altitude but is not required to meet the cooling and heating capacities of 3.6.2 and 3.6.4 at that condition

b. Temperatures ranging from 10°C (50°F) to 51.7°C (125°F)

c. Rainfall as encountered in any locale.

3.6.7 Tilted position. The air conditioner shall be capable of operating satisfactorily when its normal operating plane is at an angle of 8-1/2 degrees in any direction from the true horizontal.

3.7 Details of components

3.7.1 Air system

3.7.1.1 Fan and blower assemblies. All fan and blower assemblies shall be designed to deliver the required quantities of recirculating, ventilating, and condenser air for proper and efficient operation of the air system. The fans and blowers shall be constructed and rated in accordance with Air Moving and Conditioning Association, Inc., Bulletin 210.

3.7.1.2 Condenser blower assembly. The condenser blower assembly shall be designed for continuous operation. The blower shall deliver an adequate quantity of air, consistent with the maximum power limitation, to the condenser coil during cooling functions under extreme ambient and return air conditions.

3.7.1.2.1 Condenser inlet marking. An instructional statement shall be printed in an easily-observed location near the condenser inlet stating:

MAINTAIN 2-FOOT CONDENSER INLET CLEARANCE

Size and color of letters shall be in accordance with 3.17.

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3.7.1.3 Evaporator blower assembly. The evaporator blower assembly shall deliver sufficient air flow to meet the requirements of 3.6.3.

3.7.1.4 Air filters. Filtration of the evaporator air shall be accomplished through the use of filters in accordance with F-F-300, except the initial dust-arresting efficiency shall be not less than 60 percent at 800 fpm net face velocity. Filters provided as part of the air conditioner shall be constructed of nonferrous materials.

3.7.1.4.1 Air filter location. The evaporator air filters shall be so located that recirculated air as well as fresh air will be filtered. All air filters shall be easily accessible for removal or replacement without detaching ducts.

3.7.2 Refrigeration system

3.7.2.1 Refrigerant. The refrigeration system shall be designed for use with monochlorodifluoromethane conforming to BB-F-1421 type 22. The refrigerant system leakage rate shall not exceed 0.5 percent of the total refrigerant charge per year.

3.7.2.2 Refrigerant piping. All copper refrigerant piping shall conform to WW-T-799, type L. The tubing shall be fabricated with the use of high-temperature solder conforming to QQ-B-654.

3.7.2.3 Refrigerant drier. The size of the drier shall be such that it will have sufficient overcapacity to permit field service without requiring drier replacement.

3.7.2.4 Commercial hermetic compressor. The compressor shall have adequate cooling capacity to meet the requirements specified herein without overloading the compressor motor while operating at an ambient temperature of 51.7°C (125°F) and with return air at 51.7°C (125°F). The hermetic compressor shall be a proven commercial item and shall meet the requirements of UL Standard for Safety 984. Each compressor shall display a UL tag showing that the compressor is recognized and certified to comply with UL 984. The compressor manufacturer's standard crankcase heater shall be provided.

3.7.2.5 Condenser. The condenser shall be of an air-cooled, copper tube-aluminum fin type and shall withstand, without deformation or leakage, an internal pressure equal to 1-1/2 times the maximum working condensing pressure or the saturated refrigerant pressure at 71.1°C (160°F), whichever is greater. Condensation from the evaporator coil shall not be used to provide additional cooling effect.

3.7.2.6 Suction line accumulator. The suction line accumulator shall be of adequate size and design to prevent refrigerant liquid from returning to the compressor.

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3.7.2.7 Evaporator. The evaporator shall be of the copper tube-aluminum fin type and shall be of adequate size to limit the air velocity through the coil to 650 fpm. The evaporator shall withstand, without deformation or leakage, an internal pressure equal to 1-1/2 times the maximum working evaporation pressure or the saturated refrigerant pressure at 71.1°C (160°F), whichever is greater.

3.7.2.8 Charging valves. High and low side charging valves shall be provided for charging the refrigeration system with refrigerant without the use of special equipment. The charging valves shall be in accordance with ERDL Drawing 13216E6168.

3.7.2.9 Expansion valve. The flow of refrigerant to the evaporator coil shall be controlled by a thermostatic-type expansion valve. The valve shall be of the pressure-limiting, fully adjustable, single-ported type.

3.7.2.9.1 Inlet strainer. The expansion valve shall be equipped with an inlet strainer to prevent damage to the valve seat by foreign particles in the liquid refrigerant.

3.7.2.10 Distributor. A refrigerant distributor shall be provided as part of the evaporator coil. The refrigerant distributor shall be of the Venturi-type section.

3.7.2.11 Sight glass. A sight glass shall be provided and so located in the liquid refrigerant line from the filter-drier that the condition of the refrigerant may be observed while the air conditioner is operating and while charging refrigerant into the system.

3.7.3 Control system. All controls shall be installed in a single control panel with a weathertight cover. The control system shall be completely wired and connected for operation. Control system voltage shall be 120 volts (line to neutral).

3.7.3.1 Safety controls. Safety controls shall include the following:

a. High pressure cutout

b. Thermal protection for compressor and fan motors

c. Reset relay to prevent unit cycling on safety controls

d. Fusible plug

e. High temperature cutout for electric strip heaters

f. Circuit breaker for control circuit.

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3.7.3.2 Control panel. The control panel shall be permanently located on the air conditioner and shall be easily accessible when the ducts are attached. It shall be protected as necessary to meet the environmental exposure requirements of 3.6.6. The following devices shall be located on the control panel:

a. Control switch

b. Conditioned air thermostat

c. Circuit breaker for control circuit.

3.7.3.2.1 Control switch. Off/reset, vent and automatic cooling/heating modes shall be provided. The evaporator fan shall operate continuously in the vent and automatic cooling/heating modes.

3.7.3.2.2 Conditioned air thermostat. A cooling/heating thermostat shall be provided. There shall be a 8.3°C (15°F) temperature differential between cooling off and heating on cycles. Two step heating control shall be provided. The temperature differential shall be 1.6°C ±.55°C (3°F ±1°F). A mercury switch-type thermostat shall not be provided.

3.7.4 Electrical system

3.7.4.1 Wiring and cables. All wiring and cables used in the electrical circuit shall be firmly supported to prevent undue stress on the conductor terminals and to eliminate the possibility of a shift in conductor position. Wiring shall be protected against abrasion by means other than friction tape. Wire passing through holes in sheet metal shall be protected by grommets. Clearance between wiring and heat-generating parts shall be sufficient to prevent damage to wire insulation. Sufficient slack shall be provided to prevent stress on cable bundles, wires, and terminal connections, and to facilitate replacement of parts without disconnecting other parts. All electrical wiring shall use adequately-sized copper conductors.

3.7.4.2 Number coding. All wiring shall be number-coded. Wire 1/4-inch and under in diameter shall be coded on both sides diametrically opposite at intervals varying from 1 to 6 inches center-to-center. Wire over 1/4 inch in overall diameter shall be marked by cable markers at each detachable end.

3.7.4.3 Terminations. Before being soldered to terminals, wires shall be effectively secured to the terminals so that the connections are not dependent on soldering alone for mechanical strength. Varnish or lacquer conforming to MIL-V-173 as specified for class II materials shall be applied to the end of fibrous coverings, binders, and similar items that are not otherwise secured against fraying.

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3.7.4.4 Terminals. Terminals attached to wires may be of either the solder or solderless type, except for the following limitations on the solderless type:

3.7.4.4.1 Solderless terminals. Solderless terminals shall have a means for gripping the wire insulation. The terminals shall be used on stranded conductors ranging from 642.6 to 10380 circular mils (No. 22 to No. 10 AWG).

3.7.4.5 Relays. All relays shall hold settings for long periods of service use. The relays shall be capable of operating under the service conditions specified herein without readjusting the contact spring tension adjustment. Where a relay is required to go from the operate to nonoperate position with small current or voltage differential, the compensation for temperature effects shall be accomplished by externally connected controls.

3.7.4.5.1 Relay coils. Except for relays that are hermetically sealed, the coils used shall be vacuum impregnated with varnish conforming to MIL-V-173.

3.7.4.5.2 Relay covers. Where practicable, relays shall be covered for mechanical protection and electrical shielding to minimize radio interference. Covers shall be rugged, properly fitted, substantially dustproof, and securely fastened to prevent loosening under the service conditions specified herein. It shall be possible to reset overload relays without removing contactor box covers.

3.7.4.5.3 Time delay relay. A 1-minute time delay relay shall be included which will delay compressor operation only.

3.7.4.6 Toggle switches. Toggle switches shall be the environmentally sealed type in accordance with MIL-S-3950.

3.7.4.7 Power receptacle. An MS 90558C32412P receptacle shall be provided on the air conditioner for connecting to the appropriate end of the power cable assembly as specified in 3.7.4. The receptacle shall be mounted as flush as possible with the exterior side panel of the air conditioner cabinet.

3.7.4.7.1 Power receptacle marking. An instructional statement shall be printed in an easily-observed location near the electrical power receptacle stating:

POWER CABLE RECEPTACLE

Size and color of letters shall be in accordance with 3.17.

3.7.4.8 Power lead-in cable. The air conditioner shall contain one standard cable assembly conforming to USAF Dwg No. 72289-25 Cable Assembly - 120/208 V-3 phase-4 wire grounding, 60 Hz - 60 amps.

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3.7.5 Housing

3.7.5.1 Enclosure. The complete air conditioner shall be fully enclosed in a metallic enclosure. The enclosure shall be provided with drip channels or constructed to prevent the entry of rain or melting snow. The evaporator section shall be internally insulated to prevent condensation.

3.7.5.2 Condensation drain and pan. A condensation drain and pan shall be provided in the evaporator section to collect and dispose of moisture condensed by the evaporator coils. The drain location shall insure satisfactory condensation drainage while the air conditioner is operating.

3.7.5.2.1 Condensate drain hoses. Condensate drain hoses and all necessary fittings shall be provided to carry away condensed moisture. The hoses and fittings shall not be damaged by exposure to the temperatures as specified in 3.6.6. The hoses shall be 15 feet in length and shall be of sufficient size to carry away the condensed moisture encountered during operation at each condition of table II.

TABLE II. Dehumidifications test conditions.

Condition	Ambient Air Temperature (Dry-Bulb) °C (°F)	Return Air Temperature	
		°C (°F) (Dry-Bulb)	°C (°F) (Wet-Bulb)
1	46.1 (115)	32.2 (90)	29.4 (85)
2	46.1 (115)	26.7 (80)	23.3 (74)
3	46.1 (115)	23.9 (75)	20.6 (69)
4	46.1 (115)	21.1 (70)	18.3 (65)

3.7.5.2.2 Condensate drain marking. An instructional statement shall be printed in an easily-observed location near the condensate drain connections stating:

CONDENSATE DRAIN CONNECTION

Size and color of letters shall be in accordance with 3.17.

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3.7.5.3 Duct adapters. Each air conditioner shall be furnished with removable, insulated duct adapters. The duct adapters shall have 4-inch long by 16-inch outside diameter connections. They shall be furnished attached to the air conditioner and shall be furnished with protective end covers. When specified, grilles shall be furnished in lieu of duct adapters. (See 6.2.)

3.7.5.4 Stacking bolts. The air conditioner shall contain sufficient stacking bolts to insure that the air conditioners can satisfactorily withstand the shock test of 4.6.4.3 while three units are stacked and bolted together. Each stacking bolt shall be of the captive style using a 1-foot length of passivated stainless steel chain in accordance with RR-C-271, type II, class 3, trade number 8.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Noise levels. The noise levels produced by the air conditioner shall not exceed the sound pressure level values specified in table III when measured as specified in 4.6.5.

TABLE III. Noise level requirements.

Octave Band Center Frequencies (Hz)	Sound Pressure Level (dB ref. 0.0002 dynes/cm^2)
63	75
125	70
250	66
500	66
1000	65
2000	62
4000	54
8000	49

3.10 Dimensions. The dimensions of the air conditioner shall not exceed those stated in the drawings of 3.1.1. The overall height of the unit shall not exceed 32 inches except for the stacking bolt head.

3.10.1 Dimensional tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the air conditioner, they shall be held or limited accordingly.

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3.11 Weight. The weight of the complete air conditioner including duct adapters, refrigerant, oil, power cable, and flexible ducts shall not exceed 900 pounds.

3.12 Finishes. Finishes for application to equipment or parts thereof shall be in accordance with the following MIL-STD-808 codes for type I exposure:

Ferrous metals	F-100, pretreated with phosphate film in accordance with TT-C-490, type II
Aluminum and aluminum-alloys	F-302
Magnesium alloys	F-400
Cadmium or zinc-plated surfaces	F-221.

3.12.1 Final paint films. Final paint films shall be in accordance with MIL-C-83286. Exterior and interior coats shall be semigloss green, Color No. 24052, in accordance with FED-STD-595.

3.12.2 Primer paint films. Primer paint films shall be in accordance with MIL-P-23377.

3.13 Flexible ducts. Unless otherwise specified, (see 6.2), one 9-foot long supply air and one 7-foot long return air, 16-inch inside diameter, flexible duct in accordance with MIL-D-38386 shall be furnished with each air conditioner.

3.14 Operating instructions. Brief operating instructions and a wiring diagram of the air conditioner shall be provided in a readily-accessible location. The instructions and diagram shall be such that oil, dirt, water, et cetera, will not affect their legibility. The wiring diagram shall conform to MIL-STD-100.

3.15 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.16 Workmanship. The air conditioner, including all parts and accessories, shall be fabricated and finished in accordance with good commercial practices. Particular attention shall be given to freedom from blemishes, defects, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts; and tightness of assembly screws and bolts.

3.17 Marking. All instructions printed on the external surface of the air conditioner shall be stenciled letters and numerals that are 3 inches high with a 3/8 inch line width and shall be lusterless white, Color 37875, in accordance with FED-STD-595.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the air conditioner shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Quality conformance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Methods. Methods of testing shall be determined, instruments selected and calibrated, and tests run in accordance with ASHRAE Standard 37-60.

4.3.1.1 Positive counter. Whenever practicable, rotational speeds shall be determined by means of a directly-coupled positive counter that will count the revolutions for not less than 1 minute. A combination unit that counts the revolutions while measuring the time required in connection with other data is preferred. Strobotachometers may be used provided data can be furnished to attest to their accuracy. The use of hand tachometers shall not be acceptable.

4.3.1.2 Nozzles or orifices. Airflow shall be measured in accordance with ASHRAE Standard 37-60. Other apparatus such as pitot tubes, rotameters, et cetera, may be used only when specifically approved by the procuring activity. The rate of airflow shall be expressed in terms of cubic feet per minute.

4.3.1.3 Psychrometric chart. Humidity and enthalpy of air leaving the air conditioner shall be determined from an ASHRAE Psychrometric Chart (ASHRAE Guide and Data Book) from readings of wet- and dry-bulb temperatures. Air velocity over the wet-bulb thermometers shall be in accordance with ASHRAE Standard 37-60. The thermometers shall be so arranged that the air passing over the wet-bulb will not subsequently pass over the dry-bulb. The wick of the wet-bulb shall be kept wet with distilled water only and shall be replaced whenever a perceptible quantity of oil, dirt, incrustation, or other foreign substance has accumulated thereon.

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4.3.1.4 Mercurial barometer. Barometric pressure shall be measured by means of a mercurial barometer at the same elevation as the test. The pressure shall be corrected for the temperature of the scale and mercury and for variation in gravity resulting from the location of the barometer with regard to elevation and latitude. Readings obtained from aneroid barometers will not be acceptable.

4.3.1.5 Thermocouples and potentiometers. Temperatures shall be measured by means of appropriately located thermocouples and properly calibrated potentiometers. Insofar as practicable, thermocouples shall be insulated from contact with other metals. Temperatures shall be expressed in degrees Celsuis (Fahrenheit). Laboratory-type mercury thermometers will be permissible if used correctly in the application.

4.3.1.6 Liquid manometers. Insofar as practicable, pressure and pressure differentials shall be measured by means of liquid manometers. Pressures exceeding the range of conventional liquid manometers shall be measured by means of properly calibrated Bourdon-tube gages. Gages shall be of laboratory type having a dial diameter of not less than 4 inches.

4.3.2 Readings. Readings of all pertinent pressures, temperatures, rotational speeds, and other data necessary in demonstrating compliance with the applicable portions of Section 3 shall be recorded at intervals not exceeding 30 minutes during all tests specified herein.

4.3.3 Preliminary run-in. Any preliminary run-in specified by the manufacturer shall be performed prior to submission of the air conditioner for testing. A certified log of all preliminary running time shall be included in the air conditioner log.

4.4 Preproduction testing (see 6.2.c)

4.4.1 Test samples. Two air conditioners shall be subjected to the preproduction tests specified in 4.4.3.

4.4.2 Test report. Upon completion of the preproduction tests, a test report in accordance with MIL-STD-831 shall be prepared.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an attachment accompanying the preproduction test report (same quality and quantity of copies) or shall be included as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, cycles, or position in the test procedure as appropriate. Also, test conditions during failures or irregular operations shall be recorded.

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b. Description of the engineering reasoning behind any tests conducted to determine assignable causes for all failures and irregular operations.

c. Description of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made and reasoning behind the predicted effectiveness of those corrections.

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests specified in 4.6. The two preproduction test samples shall be allocated to the preproduction tests as follows and tested as specified in 4.6.

Sample No. 1

Examination of product	See 4.6.1
Operational tests	See 4.6.2 through 4.6.2.3
Evaporator airflow test	See 4.6.2.6
Acceleration loading tests	See 4.6.7
Endurance tests	See 4.6.3 through 4.6.3.2
Field storage and transport test	See 4.6.9
Accessory storage test	See 4.6.11
Altitude test	See 4.6.14

Sample No. 2

Examination of product	See 4.6.1
Operational tests	See 4.6.2 through 4.6.2.4
Fresh air test	See 4.6.2.5
Tilted position test	See 4.6.6

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Noise level test . . . . . See 4.6.5

Electrical test . . . . . See 4.6.8 through 4.6.8.3

Environmental tests . . . . . See 4.6.4

Heater tests . . . . . See 4.6.10

Reliability demonstration . . . . . See 4.6.12.

4.5 Quality conformance tests. The quality conformance tests shall consist of the individual tests and sampling tests.

4.5.1 Individual tests. Each air conditioner shall be subjected to the following tests as specified in 4.6:

a. Examination of product . . . . . See 4.6.1

b. Operational tests . . . . . See 4.6.2.1.2 through 4.6.2.4.

4.5.2 Sampling tests

4.5.2.1 Sampling plan. The sampling plan shall consist of testing one air conditioner out of every ten or fraction thereof produced. The air conditioner selected shall be subjected to the following tests:

a. Electrical tests . . . . . See 4.6.8

b. Heater tests . . . . . See 4.6.10.

4.6 Test methods

4.6.1 Examination of product. The air conditioner shall be inspected for the following defects to determine compliance with the requirements specified herein:

a. Critical external dimensions not as specified

b. Parts or components missing or not as specified

c. Material not as specified

d. Total weight of the unit fully charged and ready to operate exceeds the weight specified herein

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e. Parts hazardous to operating and maintenance personnel not fully enclosed, insulated, or properly guarded

f. Fans not statically and dynamically balanced

g. Connections not securely fitted to header plates

h. Panels not insulated

i. Evaporator pan and base pan leaking or not coated on the inside as specified

j. Base mounting holes not as specified

k. Undesirable control panel defects

l. Wiring requirements or wiring marking not as specified

m. Bolted and riveted connections not as specified

n. Welding defects

o. Fan motor does not indicate proper direction of rotation

p. Direction of flow not indicated on drier

q. Washers or lockwashers missing

r. Poor workmanship

s. Treatment and painting defects

t. UL tag not displayed on compressor.

4.6.2 Operational tests. The air conditioner shall be subjected to and shall satisfactorily complete the following tests. Individual records of all operational tests shall be maintained.

4.6.2.1 Cooling capacity. The preproduction air conditioner shall be tested in the cooling mode for 2 hours following stabilization in accordance with ASHRAE Standard 37-60 with controlled temperatures in accordance with condition 1 of table I. The cooling capacity shall be as specified in table I for condition 1. This test shall also be run and capacities verified at conditions 2 and 3 of table I.

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4.6.2.1.1 Production performance curves. The preproduction air conditioner shall be tested in the cooling mode with sufficient data recorded to establish five sets of performance curves for production acceptance. Prior to any data being recorded, the air conditioner shall be operated in the cooling mode for not less than 1 hour. Immediately following this run-in period, the evaporator entering-air dry-bulb temperature shall be varied from 29.4°C (85°F) to 35°C (95°F) for each performance curve required. Five sets of performance curves shall be required. Each set shall contain three performance curves: condenser air inlet temperatures of 35.0°C (95°F), 40.6°C (105°F), and 46.1°C (115°F). The five sets of performance curves (total of 15 performance curves) shall be as follows:

a. Evaporator air inlet dry-bulb temperature versus suction gage pressure

b. Evaporator air inlet dry-bulb temperature versus compressor discharge gage pressure

c. Evaporator air inlet dry-bulb temperature versus total amperage input to the air conditioner

d. Evaporator air inlet dry-bulb temperature versus total power input (kw) to the air conditioner

e. Evaporator air inlet dry-bulb temperature versus evaporator air discharge dry-bulb temperature.

Each of the performance curves shall be determined by a minimum of three points of the curve. The test conditions for each point of the curves shall be stabilized for a minimum of 15 minutes before data is recorded for plotting the curve.

4.6.2.1.2 Production cooling tests. Each production air conditioner shall be tested for 2 hours at conditions that fall within the limits of the above performance curves with entering dry-bulb temperature to the evaporator of 32.2° ±2.8°C (90° ±5°F). The results obtained on each production air conditioner will not be at variance with the performance curves recorded for the preproduction model by more than the following:

a. Suction pressure ±5 psig

b. Condensing pressure ±5 psig

c. Evaporator discharge dry-bulb temperature ±2.8°C (±5°F)

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d. Current draw ±2 amp

e. Total power consumption +0.3 kw.

4.6.2.2 Components functions. A functional test on all operating components, such as selector switches, thermostatic controls, safety controls, et cetera, shall be conducted to insure proper operation.

4.6.2.3 Leak test. A leak test shall be conducted on all components and refrigeration piping to determine the refrigerant leakage rate. A General Electric Type H-2 leak detector, or equal, shall be used to determine leaks. The combined total of all leaks shall not exceed 0.5 percent of the total refrigerant charge per year.

4.6.2.4 Stacking test. The air conditioners shall be stacked three high and bolted together using the stacking bolts as specified in 3.7.5.4. Misalignment of the stacking bolt holes or failure of the stacking bolts to fully engage the threaded fasteners of the lower unit shall be cause for rejection of the items.

4.6.2.5 Fresh air test. A test for fresh air delivery shall be performed to demonstrate compliance with 3.6.3.1.

4.6.2.6 Evaporator airflow test. Operate the air conditioner in a 46.1°C (115°F) ambient with 29.4°C (85°F) dry bulb and 21.1°C (70°F) wet bulb return air to the evaporator. Measure maximum, rated, and minimum evaporator airflow at free air delivery and 1.0 inch water gage static pressure. Evaporator airflow must meet the requirements of 3.6.3 while the duct adapters of 3.7.5.3 are attached to the unit.

4.6.3 Endurance tests. The air conditioner shall satisfactorily pass a performance test and an endurance test of 1,000 hours total duration. Once started, these tests shall not be interrupted by using the unit for any other purpose. In conducting these tests, operation of the air conditioner shall be continuous except for shutdown of the unit as necessary for servicing, changes in test conditions, minor repairs and replacement of minor parts, and for nonoperating periods associated with shift scheduling of test personnel. The total test time shall be divided as follows:

a. A 200-hour performance test on cooling and dehumidifying.

b. An 800-hour endurance test on cooling components.

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4.6.3.1 Performance tests (200 hours). The conditions for this phase of the testing shall be as specified in 4.6.3.1.1 through 4.6.3.1.3. Complete data shall be recorded during the test to show the cooling and dehumidifying performance of the air conditioner. Along with other pertinent readings, this data shall include at all operating points the wet- and dry-bulb temperatures of both the inlet and discharge evaporator air. Data shall be recorded at sufficiently frequent intervals to show the characteristics of the temperature and humidity controls until stability is reached, the length of time required to reach stability, and the ability of controls to maintain stability.

4.6.3.1.1 Cooling capacity test (170 hours). The air conditioner shall be operated with 100 percent return air for 60 hours at the first condition, 60 hours at the second condition, and 50 hours at the third condition as specified in table I. The temperature selector shall be set to insure operation in the cooling mode for each condition. The cooling output of the air conditioner shall be calculated and listed for each operating condition. In addition, it shall be shown that the maximum power required is not greater than that specified in 3.6.2. In all cases, data recording shall start at the time the air conditioner is started or the controls are reset to provide a new set of operating conditions.

4.6.3.1.2 Low ambient cooling capacity test. The air conditioner shall be operated with 100 percent return air for 10 hours as specified in table I, condition 4 to establish compliance with 3.6.2.1. In addition, the minimum ambient temperature at which the air conditioner will operate in the cooling mode shall be established.

4.6.3.1.3 Dehumidification test (20 hours). The air conditioner shall be operated with 100 percent return air for 5 hours at each of the conditions specified in table II and shall meet the requirements of 3.6.5. In all cases, data recording shall start at the time the air conditioner is started or the controls are reset to provide a new set of operating conditions. In addition, it shall be demonstrated that no entrained moisture is carried beyond the evaporator discharge of the unit. It shall also be demonstrated that no condensation accumulates on the unit casing or the duct adapter.

4.6.3.2 Endurance test (800 hours). The air conditioner shall be set up for a minimum of 1,600 test cycles or cooling and off operation. Each test cycle shall consist of 25 minutes cooling and 5 minutes off. The 1,600 test cycles shall be conducted under the following conditions:

a. 500 cycles _____ condition 1, table I

b. 500 cycles _____ condition 2, table I

c. 600 cycles _____ condition 3, table I

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At the beginning and end of the endurance test and after 1,000 operating cycles, the air conditioner shall be tested in accordance with 4.6.2.1.

4.6.4 Environmental tests. The tests specified in table IV shall be conducted in accordance with the specified procedures of MIL-STD-810. The air conditioner shall be operated prior to and after each environmental test specified. Failure of the air conditioner to start and operate normally shall be cause for rejection of the item. Component cleaning shall be limited to the use of a rag or shop cloth.

4.6.4.1 High temperature test. The highest operating temperature shall be as specified in 3.6.1.1.

4.6.4.2 Low temperature test. The lowest operating temperature shall be as specified in 3.6.1.1.

4.6.4.3 Shock test. The air conditioner shall be tested in accordance with MIL-STD-810, Method 516.2, procedure I. The shock pulse shape shall be in accordance with figure 516.2-7 (terminal peak sawtooth) or 516.2-2 (half-sine) depending on the weight with an amplitude of (b) and time duration of (d) as listed for ground equipment. Operation during the shock test is not required. The amplitude (b) shall be 40g and the time duration (d) shall be 18ms. The unit shall be tested, stacked and fastened three-high.

TABLE IV. Environmental tests.

Test Method No.	Method Title	Procedure
500.1	Low Pressure (Altitude)	I
501.1	High temperature	I
502.1	Low temperature	I
516.2	Shock	I
505.1	Solar Radiation (Sunshine)	I
506.1	Rain	I
507.1	Humidity	II
508.1	Fungus	I
509.1	Salt-fog	I
510.1	Dust (Fine Sand)	I

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4.6.5 Noise level test. The acoustic noise levels produced by the air conditioner shall not exceed the sound pressure level values specified in 3.9 at any of the octave bands or measurement positions specified in 4.6.5.2. All acoustic measurements shall be made with the air conditioner operating at maximum rated capacity. The acoustic measurement instrumentation shall be as specified in 4.6.5.3.

4.6.5.1 Background noise. Acoustic measurements shall be made only when the sound pressure levels produced by the air conditioner are more than 10 dB above the ambient background sound pressure level in such octave band.

4.6.5.2 Measurement positions. Acoustic measurements shall be made at 12 positions equally spaced every 30 degrees on a 10-foot radius circle centered on the geometrical center of the air conditioner. All measurements shall be made with the air conditioner located on level, unobstructed pavement with the microphone 5 feet above the pavement level and oriented toward the air conditioner. A flexible duct shall be attached to the outlet of the air conditioner and laid out to discharge the air away from the air conditioner.

4.6.5.3 Instrumentation. The basic required instrumentation shall consist of a sound level meter in conjunction with an octave band analyzer. The instrumentation shall meet the requirement specified in ANSI Standards S1.4 and S1.11 respectively. An up-to-date calibration and frequency response of the entire measurement system, including microphone, shall be available to permit correction of the measured data. The range of frequency response of the system shall be from 45 to 11,200 Hz and the response stated within ±2 dB. An acoustical calibrator shall be used for an end-to-end system calibration from the microphone to the meter. The sound pressure level (in dB ref: 0.0002 dynes/cm^2) produced by the calibrator, when used in conjunction with the particular microphone in the system, shall be known within 0.5 dB.

4.6.6 Tilted position tests. The air conditioner shall be tilted in four different positions, 90 degrees apart, at an angle of 8-1/2 degrees to the true horizontal plane and checked for satisfactory operation of all components during tilt. Upon conclusion of this test, the air conditioner shall be examined for damage and checked for proper operation.

4.6.7 Acceleration loading tests. The air conditioner shall be subjected to acceleration tests in accordance with MIL-A-8421. Upon completion of the flight and taxiing load tests (3g) and prior to the emergency landing load tests, the air conditioner shall be subjected to the tests specified in 4.6.3.2. The air conditioner shall be tested, stacked and fastened three high.

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4.6.8 Electrical tests. The air conditioner shall be given thorough electrical tests to determine that all circuits are inherently sound so that overall performance of the air conditioner, in compliance with the requirements of this specification, will be obtained. The electrical tests shall include, but not necessarily be restricted to, the following tests:

4.6.8.1 Continuity tests. Each electrical component and each cord and cable shall be given a continuity test to ascertain if it is wired and connected correctly and that good electrical contact is obtained.

4.6.8.2 Operating voltage tests. The operating voltage at all important points shall be checked for conformance to those shown on the circuit label and schematic drawing. This shall be accomplished with all controls set for normal operation of the air conditioner.

4.6.8.3 Safety tests. All electric motors shall be tested to insure satisfactory operation of the thermal overload protective devices in each motor. Each motor shall be tested as normally mounted in the air conditioner and while under 51.7°C (125°F) ambient temperature. The evaporator motor shall also be tested at -53.9°C (-65°F) while the compressor motor and condenser motor are tested at 10.0°C (50°F). The thermal protective devices shall be tested both by simulating a locked rotor condition and opening one of the motor phase windings during operation and prior to starting. Inability of thermal protective device to prevent damage to the motor under these abnormal conditions shall be considered a failure and require a failure analysis, a complete design re-evaluation, and applicable rework.

4.6.9 Field storage and transport test

4.6.9.1 The air conditioner shall be stacked four high without the use of shipping containers with no damage to the unit mechanically, functionally, or structurally.

4.6.9.2 The air conditioner shall be raised and lowered for four cycles with a fork lift while stacked and bolted three units high with no damage to the unit mechanically, structurally, or functionally.

4.6.9.3 Upon completion of the tests in 4.6.9.1 and 4.6.9.2, the bottom unit tested in 4.6.9.1 shall be given a capacity verification test.

4.6.10 Heater tests. The air conditioner electric strip heaters shall be inspected to insure that unsafe conditions do not exist and to insure satisfactory performance while operating as an integral part of the air conditioner. A cooling capacity test at condition 1 of table I shall be conducted prior to and subsequent to the heater tests to insure that cooling capacity performance is not adversely affected.

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4.6.10.1 Heating capacity tests. The heating capacity of the air conditioner shall be tested for conformance to the requirements of 3.6.4. The air conditioner shall be operated for 50 hours in the low heat mode and 50 hours in the high heat mode. Data recorded during the test will consist of the following:

a. Unit power consumption

b. Operating voltage

c. Operating amperage

d. Return air temperature (dry bulb)

e. Discharge air temperature (dry bulb)

f. Discharge airflow

The return air temperature will be controlled at 21.1°C ±1.1°C (70°F ±2°F) during the test. Heater capacity will be determined by the formula:

$$Q = 60 \times M \times .24 \times (T_2 - T_1)$$

where

Q = Capacity in Btu/hr

M = discharge airflow in lbs/min.

T_2 = discharge air temperature

T_1 = return air temperature.

4.6.10.2 Safety test. The evaporator fan motor shall be disconnected from the electrical circuit during this test. A cycle shall consist of the following: The heating element shall be operated in the high heat mode until the heater high temperature cutout switch activates. The evaporator compartment shall be permitted to cool until the high temperature cutout switch deactivates. The following items shall be monitored:

a. Deterioration of evaporator compartment insulation material

b. Peeling of paint on any component (interior or exterior)

c. Melting of rubber grommets or electrical wiring insulation

d. Damage to the heating elements, evaporator coil, or mounting brackets

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e. Damage to the evaporator motor or fan assembly

f. Damage to any other air conditioner component not noted above

g. Electrical leakage to ground

The heater kit shall be tested under these conditions for 10 cycles.

4.6.11 Accessory storage test. It shall be demonstrated that the supply and return ducts, the condensate drain hoses, and the power cable can be stored within the condenser section with no damage to the components or the unit. A metal box with adequate drain holes shall be attached to the condenser access door for power cable and condensate drain hose storage.

4.6.11.1 Protective end cover storage. The air conditioner shall permit storage of the duct adapter protective end covers in the condenser section without causing damage or performance degradation when the air conditioner is set up for normal operation.

4.6.12 Reliability demonstration. Satisfactory completion of all the tests specified herein without failure demonstrates compliance with the quantitative reliability requirements.

4.6.13 Maintainability demonstration. Compliance with maintainability requirements shall be demonstrated by task performances as specified below. The active maintenance down-times, the tools and equipment required, and any obstructions or conditions hindering the task or presenting potential hazards to personnel or to parts shall be observed and recorded. Signs of excessive wear or other indications of potential failure shall be recorded where observed and the information supplied to the design activity for study and recommendation of any appropriate corrections to be reflected in future items delivered. Such recommendations shall be subject to procuring activity approval before being implemented, or successful repeat of other tests specified herein may be required where the test result could be influenced by the correction. It shall be demonstrated that the item is in satisfactory operating condition after the maintenance tasks have been done. Maintenance task time data from maintenance and tear-down inspection made in connection with other specified demonstrations may be applied to the maintainability demonstration if adequate observations are made and recorded.

4.6.13.1 Corrective maintenance. Compliance with quantitative corrective maintenance down-time requirements shall be demonstrated by a maintenance demonstration procedure, maintenance task selection, and maintenance task performances in accordance with MIL-STD-471, including Appendix A and Appendix B, Method I, test plan pair (A1 plus B1).

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4.6.13.2 Preventive maintenance. Compliance with the quantitative preventive maintenance down-time requirements shall be demonstrated in accordance with MIL-STD-471, Appendix B, Method 6. Each task need not be done more than once. For maximum preventive maintenance down-time demonstration, less than 0.1 (10 percent) of the task times observed shall require greater down-time than the quantitative requirement.

4.6.14 Altitude test. The air conditioner shall be operated in the cooling mode at an altitude of 6,000 feet with an ambient temperature of 37.8°C (100°F) dry bulb and 17.2°C (63°F) wet bulb. The return air temperature entering the evaporator shall be ambient temperature. The air conditioner shall operate satisfactorily and shall not overload the compressor. The air conditioner shall also be operated in the high/low heating modes in an ambient temperature of 15.6°C (60°F) and shall operate satisfactorily during this test.

4.7 Inspection of packaging. The sampling and inspection of the preservation, packaging and interior package marking shall be in accordance with the Group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The air conditioner shall be preserved and packaged in accordance with method IIA of MIL-P-116.

5.1.2 Level C. The air conditioner shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The air conditioner, preserved and packaged as specified in 5.1.1, shall be packed in overseas-type containers conforming to PPP-B-601. The air conditioner shall be cushioned, anchored, blocked and braced in accordance with MIL-STD-1186. Skids shall be provided to facilitate machine handling. As far as practicable, containers shall be of uniform size and shape and of minimum tare and cube. Closure and strapping shall be in accordance with the appendix of the specification.

5.2.2 Level B. Level B packing shall be the same as level A, except the containers shall be of the domestic type of PPP-B-601. Skids shall be provided to facilitate machine handling.

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5.2.3 Level C. The air conditioner preserved and packaged as specified in 5.1.2 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Air Conditioner A/E32C-39

6. NOTES

6.1 Intended use. The A/E32C-39 air conditioner is intended for use in cooling, heating, dehumidifying, filtering, and circulating air for portable shelters or vans in accordance with electronic and personnel requirements. The air conditioner is designed to be used while being directly exposed to the atmosphere.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. In the event it becomes necessary to change said drawings in order to achieve the performance requirements of this specification, the contractor shall immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved, such changes will be processed in accordance with the provisions of the changes clause of the contract (see 3.1.1).

c. Location and conditions for preproduction testing (see 4.4).

d. When supply and return grilles are required and the duct adapters are not required except for testing (see 3.7.5.3 and 4.6.2.5).

e. When external flexible ducts are not to be provided (see 3.13 and 4.6.5.2).

f. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

g. Special shipment marking (see 5.4).

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6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent

$$MTBF = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \text{ where } x = \frac{\text{Mission time}}{MTBF}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degrees freedom) distribution even though the failure rate for the test time is zero.

6.3.2 Mean-time-between-failures. The mean-time-between-failures (MTBF) is the arithmetical mean (average) of the operating time between failures.

6.4 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

ac . . . . . alternating current

amp . . . . . ampere

ASA . . . . . American Standards Association

ASHRAE . . . . . American Society of Heating, Refrigeration, and Air Conditioning Engineers, Inc.

AWG . . . . . American wire gage

BTU/hr . . . . . British thermal units per hour

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cfm . . . . . cubic feet per minute

fpm . . . . . feet per minute

g . . . . . gravitational units

Hg . . . . . mercury

in wg . . . . . inches water gage

kw . . . . . kilowatt

MTBF . . . . . mean-time-between-failure

V . . . . . volt

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Review activities:
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